

10.1 Functions of two or more variables

Domain and range

polynomials: ex. $x^2 + 3xy + 4y^2$

no restriction

Domain: $\mathbb{R}^2$ logarithms: $\ln x$, $x \in (0, \infty)$ $\ln(y - x^2)$, $\{(x, y) \in \mathbb{R}^2 \mid y - x^2 > 0\}$ $\ln(3xy - 4y^2)$, $\{(x, y) \in \mathbb{R}^2 \mid 3xy - 4y^2 > 0\}$ rational functions $\frac{1}{(x+3)(x-2)}$ $\{x \in \mathbb{R} \mid x-3 \neq 0, x-2 \neq 0\}$ $\frac{1}{4xy - 3x^2}$ $\{(x, y) \in \mathbb{R}^2 \mid 4xy - 3x^2 \neq 0\}$ $\frac{1}{3x^2 + 2y^2}$ $\{(x, y) \in \mathbb{R}^2 \mid (x, y) \neq (0, 0)\}$ square roots $\sqrt{x^2 - 2}$ $\{x \in \mathbb{R} \mid x^2 - 2 \geq 0\}$ $\sqrt{x^2y + 4xy}$ $\{(x, y) \in \mathbb{R}^2 \mid x^2y + 4xy \geq 0\}$

square roots $\sqrt{x^2-2}$ $\{x \in \mathbb{R} \mid x^2-2 \geq 0\}$

$$\sqrt{x^2y+4xy} \quad \{(x,y) \in \mathbb{R}^2 \mid x^2y+4xy \geq 0\}$$

$$\frac{1}{\ln x^2+y^2}$$

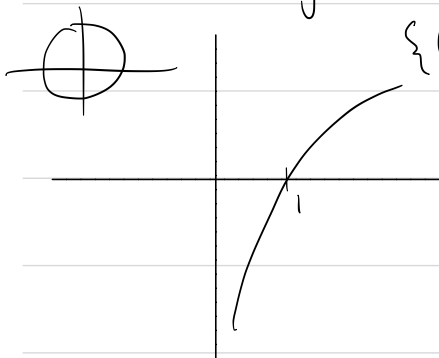
$$x^2+y^2 > 0$$

$$\ln(x^2+y^2) \neq 0$$

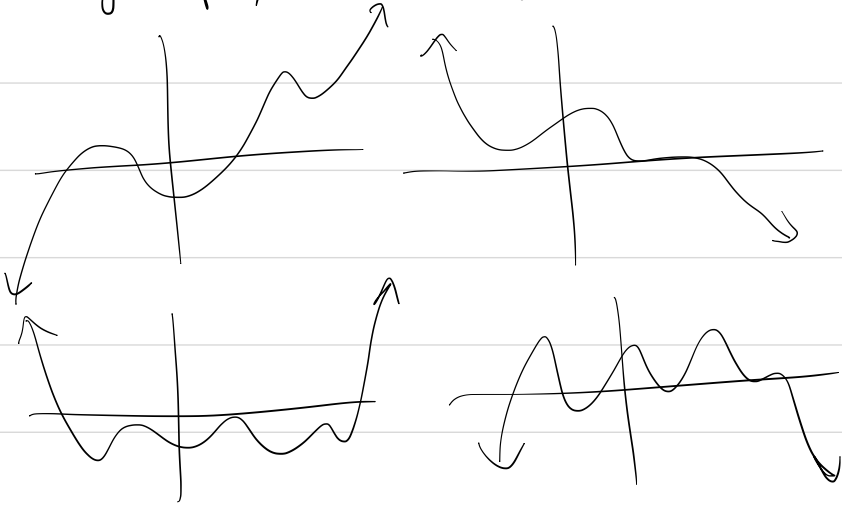
$$\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \neq 1, \\ x^2+y^2 \neq 0\}$$

$$(-\infty, \infty) \times (-\infty, \infty)$$

$$\mathbb{R} \times \mathbb{R} \quad (x,y)$$



Range: polynomial "Range"



$$x^2 + 3y^2$$

$$ax^2 + by^2$$

$$\in \mathbb{R}$$

$$x^4 + 3x^2y^6 \geq 0$$

$$\ln(x) \quad \begin{array}{c} \nearrow \\ \searrow \end{array} \quad (-\infty, \infty) = \mathbb{R}$$

$$\sqrt{x^2} \quad (0, \infty)$$

$$\frac{1}{x}$$

$$(-\infty, 0) \cup (0, \infty) = \mathbb{R} \setminus \{0\}$$

$$\{x \in \mathbb{R} \mid x \neq 0\}$$

$$\frac{1}{\ln(x^2+y^2)}$$

$$x^2+y^2 \neq 1 \quad x^2+y^2 \neq 0$$

$$(-\infty, 0) \cup (0, \infty)$$

$$\text{Ran}[\ln(x^2+y^2)] = (-\infty, 0) \cup (0, \infty)$$

$$\text{Ran}\left[\frac{1}{\ln(x^2+y^2)}\right] = (-\infty, 0) \cup (0, \infty)$$

$x > 0$
Dom

$$\ln(y - x^2) \quad \text{Dom: } \{(x, y) \in \mathbb{R}^2 \mid y - x^2 > 0\}$$

$$y > x^2$$



$$\text{Ran: } (-\infty, \infty)$$

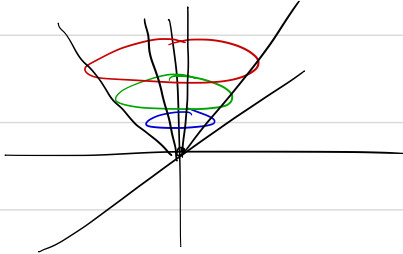
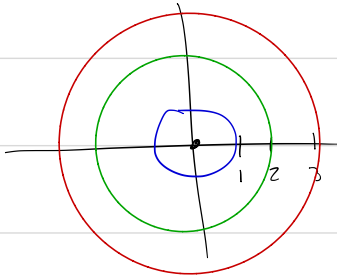
$$\text{Ran } \{y > x^2\} = \{z \in \mathbb{R} \mid z > 0\}$$



$$f(x, y) = \sqrt{x^2 + y^2}$$

$$\underline{z} = \sqrt{x^2 + y^2}$$

$$c = \sqrt{x^2 + y^2}$$



$$\text{Dom: } \mathbb{R}^2$$

$$\text{Ran: } \{z \in \mathbb{R} \mid z \geq 0\}$$

$$(0, \infty)$$

$$c = 0, 1, 2, 3 \in \text{Ran}(f)$$

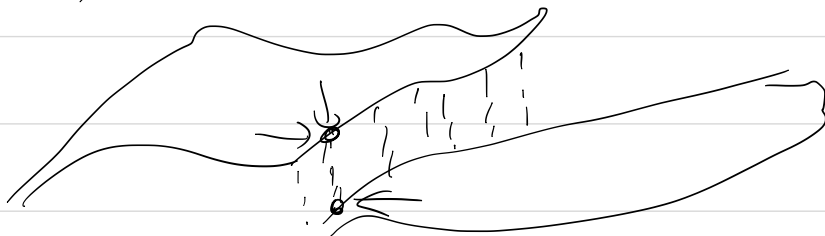
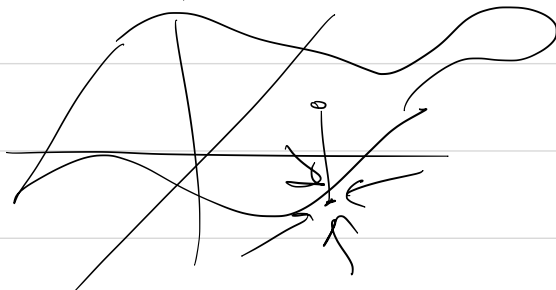
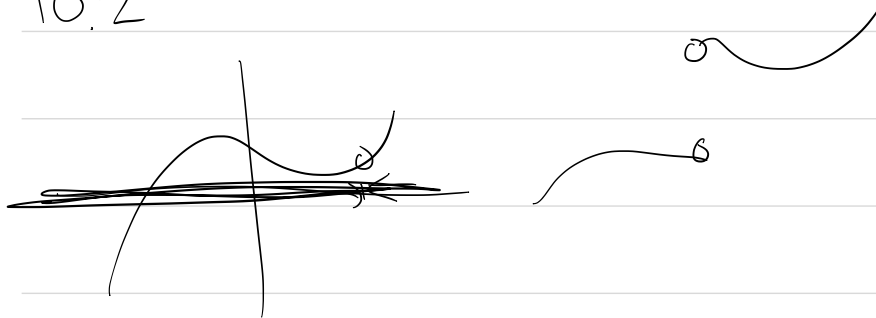
$$- 0 = \sqrt{x^2 + y^2}$$

$$- 1 = \sqrt{x^2 + y^2} \quad x^2 + y^2 = 1$$

$$- 2 = \sqrt{x^2 + y^2} \quad x^2 + y^2 = 4$$

$$- 3 = \sqrt{x^2 + y^2} \quad x^2 + y^2 = 9$$

10.2



$$y = x$$

$$y = x^2$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - 2y^2}{x^2 + y^2}$$

① positive x-axis $\rightarrow y=0$

② positive y-axis $\rightarrow x=0$

$$\textcircled{1} \lim_{x \rightarrow 0^+} \frac{x^2 - 2(0)^2}{x^2 + (0)^2} = \lim_{x \rightarrow 0^+} \frac{x^2}{x^2} = \lim_{x \rightarrow 0^+} 1 = 1$$

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$$\textcircled{2} \lim_{y \rightarrow 0^+} \frac{0^2 - 2y^2}{0^2 + y^2} = \lim_{y \rightarrow 0^+} \frac{-2y^2}{y^2} = \lim_{y \rightarrow 0^+} -2 = -2$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - 2y^2}{x^2 + y^2} = \text{DNE}$$

